NITTE | NMAM INSTITUTE OF TECHNOLOGY

CODE FOR

INNOVEX

RESQNET: OFFLINE SOS NETWORK

TRACK NAME: DISASTER RESILIENCE (PS2)

TEAM NAME:

TEAM ID:

TEAM MEMBERS:



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THE PROBLEM

- **The situation:** During floods, cyclones, tsunamis and landslides, mobile networks, internet connectivity and power supply often fail together. Victims are left with no way to call for help.
- **The gap:** Trapped people cannot share their exact location, their needs or how many of them are stranded. Rescue teams receive scattered, unprioritized reports and cannot tell where to go first.
- **Who faces it:** Residents of flood-prone, coastal and hill regions of India, along with rescue teams and volunteers, during every monsoon and cyclone season.
- **Our track:** Disaster Resilience (PS2): faster coordination, search and rescue, and offline-first communication during a crisis.

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THE SOLUTION

- **What we are building:** RESQNET, an offline-first SOS system. A victim's Android phone sends GPS location and emergency details to a server on a rescue-team laptop, which shows every request on a command-center dashboard.
- **How it works:** The phone sends one structured SOS packet (type, people, injuries, GPS) to the rescue team's hotspot. Out of range, it stays queued and sends itself when the app finds the hotspot. The laptop confirms receipt. No internet needed.
- **How it solves the problem:** A visible urgency score (3 x children + 2 x injured + type weight + minutes waiting) ranks every SOS, likely duplicates are flagged for merging, and each case moves from Pending to Dispatched to Rescued.
- **What makes it different:** Bridgefy, Briar and Meshtastic are general offline messengers. RESQNET delivers a structured, prioritized, rescue-ready SOS, and the rescue team brings the network to the victims.

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TECH & FEASIBILITY

1

Android app (Kotlin)

Builds: SOS form (type, people, injuries), GPS and an offline queue. **Output:** a saved SOS packet, even with no signal.

2

Server (Python Flask + SQLite)

Builds: receives each SOS, scores urgency, flags duplicates. **Output:** one ranked case list for the operator.

3

Local Wi-Fi hotspot link

Builds: the app scans for the rescue hotspot and sends on connect. **Output:** SOS delivered within about 100 m, receipt confirmed.

4

Rescue dashboard

Builds: ranked list with GPS positions and case status. **Output:** the team knows where to go first.

5

Demo and testing

Builds: one laptop, one Android phone, free tools. **Output:** a live test of send, queue, reconnect and rank, with range measured.

6

Range and limits

Limit: about 100 m direct range, less through rubble; it cannot raise the first alert. **Planned:** phone relay and LoRa for range.

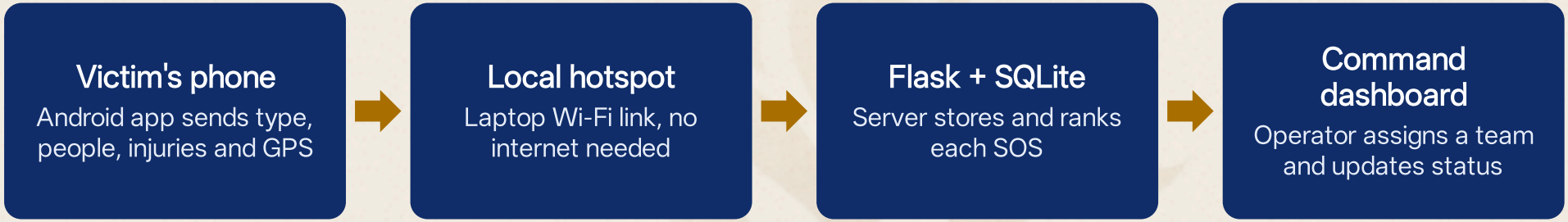
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IMPACT

- **Who benefits:** People in flood, cyclone and landslide-prone areas, rescue teams, NSS volunteers and local administration.
- **Social impact:** Rescue teams that reach a disaster zone find and rank survivors in minutes instead of searching blind, so children and the injured are reached first.
- **Economic impact:** Costs almost nothing: existing Android phones, one laptop at the rescue point and free open-source software, with no new towers.
- **Theme:** Navyug – A New Era. Technology that keeps communities connected when infrastructure fails.
- **Growth beyond the hackathon:** Phone-to-phone relay, LoRa gateways and drone-mounted receivers to explore for longer range, solar charging, voice SOS in Kannada and Hindi, signed reports to block false SOS, and district-level pilots.

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ARCHITECTURE & DEMO



Live demo plan

1. Switch off mobile data and internet on the phone.
2. Connect the phone to the laptop's local hotspot.
3. Submit an SOS with the emergency type, number of people, injuries and GPS location.
4. Go out of range, submit an SOS: it queues, sends on reconnect, laptop confirms.
5. Show the request, its GPS location and urgency rank on the dashboard, then update its status.

GitHub repository: add link here | **Demo video (unlisted, 1 to 2 min):** add link here

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